TRISTAN ALDERSON

Work Authorization: Dual US & Canadian Citizen

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EDUCATION

Bachelor of Applied Science in Electrical and Computer Engineering — Kingston, ON, Canada
Queen's University • *Dean's Scholar* — Graduation: 05/2026

PROFESSIONAL PROFILE

Hardware-aware systems engineer specializing in bridging the gap between custom PCB architectures and low-level Linux/RTOS execution environments. Experienced in full-stack board bring-up, custom bootloader configurations, writing deterministic drivers, and utilizing bench tools to isolate transient electrical and logical faults across the hardware-software boundary.

TECHNICAL SKILLS

Languages:	C, C++, Python, Bash, MATLAB, Verilog, SystemVerilog
Embedded Linux:	Buildroot, U-Boot, Device Trees, Rootfs/Image Integration, GDB, Jenkins CI/CD
Hardware & Bring-Up:	KiCad, LTspice, PCB Multi-layer Layout, Component Selection, Oscilloscopes, Logic Analyzers
Protocols & Interfaces:	CAN, SPI, I2C, UART, AXI4, AXI DMA, TCP/IP, UDP, Ethernet, NTP/PTP, 1G/10G/100G SFP/QSFP
Power & RF Design:	DC-DC Buck Regulators, PMOSFET Protection Circuits, Zener Clamping, Controlled Impedance RF Layout
Target Platforms:	ARM Cortex-A53 Linux, STM32 (Cortex-M), ESP32, AMD Xilinx Kria KV260, Marvell Switch ASICs

PROFESSIONAL EXPERIENCE

Embedded Software Engineer Intern — 05/2024 – 08/2025
Evertz Microsystems — Burlington, ON, Canada

- Architected high-performance **C++ Linux platform control services** for a distributed, multi-node 1RU embedded chassis, utilizing **gRPC**, **REST APIs**, and **Redis-backed architecture** to safely synchronize inter-node telemetry.
- Designed low-level, hardware-facing protocols for deterministic **UDP stream telemetry** and reboot-safe GNSS binary state preservation across multiple hot-swappable computing elements.
- Isolated and resolved high-speed hardware initialization and link-stability failures on **100G QSFP data paths** by cross-referencing **GDB** thread states with physical multi-layer board traces and vendor SDK metrics.
- Automated validation workflows by optimizing hardware-upload script infrastructure and upgrading core communication layers utilizing OpenSSL within custom **Buildroot** target environments.

ENGINEERING DESIGN TEAMS & PROJECTS

Avionics Lead — 05/2025 – Present
Queen's Rocket Engineering Team (QRET) — Kingston, ON, Canada

- Direct a 5-person engineering sub-team designing distributed hardware and real-time firmware for rocketry valve actuation nodes, **ESP32 telemetry boards**, **CAN communication networks**, and safety-critical launch interfaces.
- Developed **AimNetwork**, a low-latency, fixed-format **CAN bus protocol** featuring 64-bit payloads, rigid 3-bit addressing topologies, structured node heartbeat checks, and fault-tolerant WiFi/LoRa link forwarding layers.
- Authored zero-dynamic-memory, hardware-abstracted **STM32 CAN drivers** with strict 16-frame TX/RX bounded queues; verified 100% communication packet retention under heavy 90% bus-load injection simulations.
- Executed comprehensive board-level bring-up and bench validation of custom flight microcontrollers; tracked and mitigated a thermal fault in a zener-clamped power distribution circuit via manual rework and parameter adjustment.

LiteLM: FPGA-Backed LLM Inference on ARM Linux — 08/2025 – 03/2026
Queen's University Showcase Project — Kingston, ON, Canada

- Engineered a hybrid compute subsystem on an **AMD Xilinx Kria KV260 SoM**, splitting parallel matrix math offloads to custom programmable logic while maintaining sequential execution on an **ARM Cortex-A53 Linux platform**.
- Built the deterministic hardware-software boundary by mapping explicit **AXI4 DMA** physical addresses, configuring custom Linux Device Trees, managing U-Boot overlays, and orchestrating zero-copy **udmabuf** kernel blocks.
- Awarded **3rd Place for Top Computer Engineering Project** across the entire departmental ECE Showcase.

AI Engineer & Avionics Developer — 10/2023 – 08/2024
Queen's Rocket Engineering Team (QRET) — Kingston, ON, Canada

- Modeled, fabricated, and validated a modular **SAM-M10Q GPS daughter board** leveraging precise I2C data isolation and a maximized copper ground plane to suppress high-frequency RF noise and preserve satellite lock during flight profiles.
- Secured **2nd Place Overall at the Launch Canada Competition** through the successful implementation of real-time rocket video tracking using YOLOv8, integrated with MATLAB-simulated dual-axis PID tracking mounts.